

PAYNESVILLE, MN, USA

WMO: 720367

Lat: 45.372N

Lon: 94.746W

Elev: 1180

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-11.8	-23.6	1.6	-16.3	-18.1	2.1	-11.0	24.9	13.6	23.0	15.5	4.9	280	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	87.9	71.5	84.4	70.0	82.1	68.2	75.7	83.8	73.5	80.8	71.5	78.8	10.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	128.2	80.8	71.5	121.5	79.0	69.4	113.1	76.2	40.1	83.3	37.9	81.1	36.1	78.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	19.5	17.4	-24.4	92.7	5.4	4.0	-28.3	95.6	-31.4	98.0	-34.4	100.2	-38.4	103.2
			-24.7	78.6	5.3	2.0	-28.5	80.0	-31.6	81.2	-34.6	82.4	-38.4	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	11.8	14.5	30.0	43.0	56.3	65.6	70.3	67.3	60.1	45.9	31.0	17.7	(6)
	DBStd	22.90	14.10	12.59	12.68	10.37	9.05	6.22	5.89	5.43	8.48	9.34	11.23	12.24	(7)
	HDD50	4878	1185	994	628	251	38	0	0	0	13	194	573	1002	(8)
	HDD65	8503	1650	1414	1084	659	300	65	19	37	193	596	1019	1467	(9)
	CDD50	2305	0	0	9	42	232	469	628	537	317	67	4	0	(10)
	CDD65	455	0	0	0	0	29	84	182	109	47	4	0	0	(11)
	CDH74	4268	0	0	0	13	368	758	1748	920	415	46	0	0	(12)
	CDH80	1122	0	0	0	2	115	183	501	212	102	7	0	0	(13)
(14) Wind	WSAvg	7.7	8.0	8.1	8.3	9.1	8.4	7.2	6.1	5.8	7.1	8.2	8.4	7.8	(14)
(15) Precipitation	PrecAvg	30.1	0.8	0.8	1.7	2.8	3.3	4.8	3.9	3.7	3.2	2.3	1.5	0.7	(15)
	PrecMax	38.4	2.0	2.0	3.5	7.0	6.7	8.0	7.4	7.6	8.2	4.8	4.0	1.7	(16)
	PrecMin	23.4	0.0	0.1	0.6	0.8	0.9	2.0	0.9	0.4	0.5	0.7	0.0	0.3	(17)
	PrecStd	4.6	0.6	0.5	0.8	1.5	1.5	1.5	1.7	1.6	1.9	1.2	1.3	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	50.9	68.0	75.1	90.0	88.7	92.8	88.5	87.7	80.6	63.4	45.8	(19)
		MCWB	36.3	46.7	57.2	57.3	65.7	71.5	76.4	72.5	71.1	62.1	52.8	41.2	(20)
	2%	DB	36.7	40.1	60.6	70.0	82.3	84.4	88.3	84.8	82.8	72.5	56.8	40.1	(21)
		MCWB	34.1	36.6	52.6	53.7	64.0	68.8	73.2	71.9	68.5	58.7	47.4	36.5	(22)
	5%	DB	34.2	36.0	53.7	65.1	77.4	81.5	84.7	82.0	79.1	66.1	51.7	36.3	(23)
		MCWB	32.7	33.6	48.8	51.2	60.7	66.4	71.4	68.9	66.4	56.0	44.8	34.3	(24)
	10%	DB	31.7	32.2	46.7	60.7	72.8	78.6	82.1	79.3	74.9	61.1	46.1	33.7	(25)
		MCWB	30.8	30.5	42.9	48.7	59.3	65.1	69.5	67.7	65.1	52.4	41.4	32.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	47.0	60.0	59.3	71.8	76.3	79.5	76.6	75.6	67.2	55.9	42.3	(27)
		MCDB	40.9	52.7	65.4	72.0	84.2	84.2	86.4	84.4	83.9	73.4	63.2	44.6	(28)
	2%	WB	34.5	37.1	53.6	55.9	67.8	72.7	76.9	74.4	72.0	61.0	49.8	36.9	(29)
		MCDB	36.4	39.6	57.1	65.4	77.5	80.2	85.5	81.7	77.9	68.6	53.8	39.3	(30)
	5%	WB	32.6	33.5	47.6	53.1	64.2	69.5	74.4	72.4	69.4	57.2	45.7	34.4	(31)
		MCDB	34.0	35.7	52.7	61.7	72.1	77.0	82.4	79.0	75.8	65.0	50.3	35.9	(32)
	10%	WB	30.4	30.6	42.6	50.2	61.3	67.4	72.0	70.4	66.5	53.4	41.7	32.5	(33)
		MCDB	31.4	32.1	47.0	58.9	69.5	74.8	79.6	76.4	72.7	58.8	46.4	33.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.3	16.7	19.4	21.5	19.9	21.1	20.5	21.0	18.6	16.4	15.1	(35)
		MCDBR	17.6	19.9	24.5	28.8	29.3	23.7	22.3	23.2	25.2	26.1	22.2	15.9	(36)
	5% WB	MCWBR	15.1	16.4	16.0	15.4	13.4	11.2	10.6	11.6	12.5	14.5	14.5	13.6	(37)
		MCWBR	16.8	18.7	22.2	23.2	22.6	19.9	19.4	19.6	20.3	22.0	19.6	14.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.260	0.277	0.308	0.363	0.391	0.387	0.385	0.398	0.360	0.312	0.281	0.255	(40)
		taud	2.426	2.371	2.402	2.316	2.299	2.358	2.375	2.327	2.414	2.535	2.521	2.456	(41)
	Ebn at Noon		276	290	294	284	279	279	278	270	272	274	266	266	(42)
		Edh at Noon	22	28	32	38	40	38	37	37	32	24	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		552	895	1203	1490	1715	1905	2043	1699	1304	828	530	387	(44)
	RadStd		70	102	110	129	132	128	72	139	97	118	51	48	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page